

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Physics — SSC-I (9th Grade)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

General Instructions:

1. Write your name, roll number and section clearly on the answer sheet before starting.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt **all 11** questions. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt **all 4** questions. For each question, attempt either the main question **OR** its alternative.
5. Use $g = 10 \text{ m/s}^2$ in numerical problems unless otherwise specified. Scientific calculator allowed.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

#	Question	A	B	C	D	Answer
1	Which measuring instrument is most suitable for finding the duration of an event to a precision of 0.1 s?	Digital stopwatch	Analogue stopwatch	Metre rule	Physical balance	ⒶⒷⒸⒹ
2	A vernier caliper has a main scale with smallest division 0.5 mm and a vernier scale with 25 divisions. Its least count is:	0.05 mm	0.02 mm	0.1 mm	0.2 mm	ⒶⒷⒸⒹ
3	A motorboat increases its speed uniformly from 8 m/s to 26 m/s in 6 s. Its acceleration is:	34 m/s ²	18 m/s ²	3 m/s ²	4.33 m/s ²	ⒶⒷⒸⒹ
4	A ball is thrown vertically upward with a speed of 25 m/s. Taking $g = 10 \text{ m/s}^2$, the time taken to reach maximum height is:	1.25 s	5 s	25 s	2.5 s	ⒶⒷⒸⒹ
5	A net force of 32 N acts on a shopping trolley of mass 8 kg. Its acceleration is:	4 m/s ²	0.25 m/s ²	40 m/s ²	256 m/s ²	ⒶⒷⒸⒹ
6	In the free-body diagram (FBD) of a book resting on a horizontal table, which forces on the book are shown?	Only its weight	Weight, Normal force, and friction	Weight and the Normal force from the table	Only the Normal force	ⒶⒷⒸⒹ
7	For an L-shaped (bent) uniform lamina, the centre of gravity:	Always lies within the material of the lamina	Always lies exactly at the bend	Cannot be determined for irregular shapes	May lie outside the lamina, in empty space	ⒶⒷⒸⒹ
8	A force of 25 N is applied perpendicular to a spanner at a distance of 0.3 m from a nut. The torque produced is:	7.5 N·m	25.3 N·m	0.012 N·m	75 N·m	ⒶⒷⒸⒹ
9	Taking the density of mercury as 13,600 kg/m ³ and $g = 10 \text{ m/s}^2$, the pressure exerted by a 0.76 m column of mercury is closest to:	$1.79 \times 10^3 \text{ Pa}$	$1.03 \times 10^5 \text{ Pa}$	$1.79 \times 10^4 \text{ Pa}$	$1.03 \times 10^6 \text{ Pa}$	ⒶⒷⒸⒹ
10	A pump does 3000 J of work in 20 s. Its power output is:	60 W	6667 W	150 W	15,000 W	ⒶⒷⒸⒹ
11	Which renewable energy resource harnesses heat trapped within the Earth's crust from the decay of radioactive elements?	Geothermal	Biomass	Tidal	Wave	ⒶⒷⒸⒹ
12	Which branch of physics is most concerned with studying the dynamics of blood flow and respiration in the human body?	Astrophysics	Geophysics	Acoustics	Biophysics	ⒶⒷⒸⒹ

SECTION B — Short Questions ($11 \times 3 = 33$ Marks)

Attempt all 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	OR Question	Marks
2(i)	Describe the construction of a physical balance (beam, pans, pointer, levelling screws) and explain how it is used to measure the mass of a small object.	3	OR	Describe how a measuring cylinder is used to find the volume of an irregular solid object. A stone is lowered into a measuring cylinder and the water level rises from 24 mL to 63 mL. Calculate the volume of the stone.	3
2(ii)	Write the following measurements in scientific notation: (a) 0.000085 m, (b) 52,000,000 s. Express the value in (a) using an appropriate SI prefix.	3	OR	Write the SI prefix and symbol for each of the following multipliers: 10^{-2} , 10^6 , 10^9 . Then convert 250 mm into metres, expressing your answer in scientific notation.	3
2(iii)	A car's speed increases uniformly from 6 m/s to 26 m/s in 8 s. Calculate (a) its acceleration, (b) the distance travelled using $s = ut + \frac{1}{2}at^2$.	3	OR	Differentiate between uniform and non-uniform acceleration, with one example of each. A train decelerates uniformly from 40 m/s to rest in 8 s. Calculate its deceleration.	3
2(iv)	A jogger runs 500 m along a straight road and then jogs back 200 m towards a bench. Calculate (a) the total distance covered, (b) the magnitude	3	OR	What information is given by the slope of a speed-time graph, and by the area under it? A car travels at a constant speed of 15 m/s for 8 s.	3

	of her displacement from the starting point.			What distance does it cover, as read from the area under its speed-time graph?	
2(v)	State Newton's second law of motion and write its mathematical form $F = ma$. A supermarket trolley of mass 7 kg is pushed with a net force of 42 N. (a) Calculate its acceleration. (b) If the same force were applied to an empty trolley of mass 3.5 kg instead, what would the new acceleration be?	3	OR	Define inertia. Using the concept of inertia, explain why dust particles fly off a hanging carpet when it is beaten with a stick.	3
2(vi)	A hockey stick strikes a ball of mass 0.16 kg, changing its speed from rest to 30 m/s in 0.02 s. Calculate (a) the change in momentum of the ball, (b) the average force exerted by the stick.	3	OR	State the law of conservation of momentum. A 45 kg boy standing at rest on a skateboard throws a 3 kg bag horizontally at 6 m/s. Calculate (a) the boy's recoil speed, (b) his kinetic energy immediately after throwing the bag.	3
2(vii)	A see-saw is pivoted at its centre. Rabia (weight 300 N) sits 0.8 m from the pivot. At what distance from the pivot must Adeel (weight 400 N) sit on the other side to balance the see-saw?	3	OR	State the principle of moments. A uniform metre rule is pivoted at the 40 cm mark. A weight of 9 N hangs at the 10 cm mark. At what mark on the other side must a 6 N weight be hung to balance the rule (ignore the weight of the rule)?	3
2(viii)	Explain why a bus with a high centre of gravity is more likely to overturn while rounding a sharp bend than a car with a low centre of gravity. Relate your answer to the position of the centre of gravity and stability.	3	OR	What is meant by the centre of gravity of a body? Describe, with the help of a diagram, how the centre of gravity of an irregularly shaped lamina (thin sheet) can be found experimentally using a plumb line.	3
2(ix)	What is meant by a dissipative force? Using the example of a meteor entering the Earth's atmosphere, explain how friction causes energy to be dissipated as heat.	3	OR	Explain, with reference to friction, why rubbing your hands together makes them feel warm on a cold day. Name the type of energy conversion that takes place.	3
2(x)	A tank is filled with kerosene oil of density 800 kg/m ³ to a depth of 4 m. Calculate the pressure exerted by the oil at the bottom of the tank. ($g = 10 \text{ m/s}^2$)	3	OR	Define a manometer and state its working principle. A U-tube manometer containing a liquid of density 900 kg/m ³ shows a height difference of 25 cm between its two limbs. Calculate the pressure difference it indicates. ($g = 10 \text{ m/s}^2$)	3
2(xi)	Define pressure and state its SI unit. A tractor of mass 2500 kg has a total tyre contact area of 0.5 m ² with the ground. Calculate the pressure it exerts on the soil. ($g = 10 \text{ m/s}^2$)	3	OR	Using the concept $P = F/A$, explain why a tractor is fitted with wide tyres while a nail has a sharp pointed tip. Give one more real-life example of increasing or decreasing pressure by changing the area of contact.	3

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	OR Question	Marks
3(i)	(a) Derive the equation of motion $s = ut + \frac{1}{2}at^2$ using a speed-time graph. [2] (b) A car moving at 12 m/s accelerates uniformly at 2 m/s ² for 6 s. Find (i) its velocity after 6 s, (ii) the distance covered in this time using the derived equation. [2] (c) Verify your answer to (b)(i) using $v^2 = u^2 + 2as$. [1]	5	OR	(a) State the three equations of motion and define each symbol used. [2] (b) A stone is dropped from the top of a tower and takes 5 s to hit the ground ($g = 10 \text{ m/s}^2$, ignore air resistance). Calculate (i) the height of the tower, (ii) the velocity with which it strikes the ground. [3]	5
3(ii)	(a) State the law of conservation of momentum and write its mathematical form for a two-body collision. [1] (b) A 1000 kg car moving at 15 m/s collides with a stationary 500 kg car, and the two vehicles lock together on impact. Calculate their common velocity just after the collision. [2] (c) Calculate the loss in total kinetic energy during the collision. [2]	5	OR	(a) Define impulse and derive the impulse-momentum relation $F = \Delta p / \Delta t$ from Newton's second law. [2] (b) A karate expert's striking hand and forearm (effective mass 1.2 kg) hits a wooden board, decelerating from 12 m/s to rest in 0.008 s. Calculate (i) the impulse delivered, (ii) the average force exerted by the board on her hand. [3]	5
3(iii)	(a) Derive the expression $P = \rho gh$ for the pressure exerted by a liquid column of height h and density ρ . [2] (b) A tank contains fresh water (density 1000 kg/m ³) filled to a depth of 2.5 m, with a 0.5 m thick layer of oil (density 800 kg/m ³) floating on top. Taking $g = 10 \text{ m/s}^2$, calculate (i) the pressure at the oil-water interface due to the oil layer alone, (ii) the total pressure at the bottom of the tank due to both liquids (ignore atmospheric pressure). [3]	5	OR	(a) Describe the construction and working of a simple mercury barometer, explaining what the height of the mercury column represents. [2] (b) At a hill station, the atmospheric pressure is measured as a mercury column of height 0.70 m (density of mercury = 13,600 kg/m ³ , $g = 10 \text{ m/s}^2$). Calculate the atmospheric pressure in pascals at that location, and state whether this is higher or lower than the sea-level value of $1.013 \times 10^5 \text{ Pa}$. [3]	5
3(iv)	(a) State the law of conservation of energy. [1] (b) A 4 kg stone is thrown vertically upward from the ground with a speed of 20 m/s. Taking $g = 10 \text{ m/s}^2$: (i) Calculate its kinetic energy at the point of projection. (ii) Using conservation of energy, calculate the maximum height it would reach if there were no air resistance. [3] (c) If air resistance limits the stone to an actual maximum height of 18 m, calculate the work done against air resistance. [1]	5	OR	(a) Differentiate between renewable and non-renewable energy resources, giving two examples of each (other than fossil fuels, e.g. geothermal, tidal, biomass, wind). [2] (b) A small hydroelectric turbine allows 2000 kg of water to fall through a height of 15 m every second. Taking $g = 10 \text{ m/s}^2$, calculate (i) the gravitational potential energy lost per second by the falling water, (ii) the power output of the generator if all of this energy is converted to electrical energy each second. [3]	5

Invigilator's Signature: _____

Candidate's Signature: _____

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— End of Question Paper —